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Study of physico-chemical factors of two ponds Taj Baj Pokhra and Rani Pokhar of Vaishali District, Bihar

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Abstract- Fresh water is essential for the existence of life. The quality of water depends upon physico-chemical and biological characteristics which reflect on biotic status of the ecosystem. The present study evaluated the physico-chemical characteristics of the waters in two fish ponds, located in Vaishali, Bihar. For surface water determination of water quality index becomes essential and pre-requisite. Analysis of some physico-chemical characteristics like water temperature, total dissolved solids, pH, dissolved oxygen, BOD, total alkalinity, total hardness has been done during the investigation period. The physicochemical parameters were recorded by using standard methods for a period of one year (March, 2022 to February 23). The dissolved oxygen was found in the range of 6.23 mg/L to 7.80 mg/L. The alkalinity was in the range of 31.8 mg/L to 53.10 mg/L. The temperature was in the range of 16.3°C to 32.4°C. The hardness was in the range of 185 mg/L to 245 mg/L. Seasonal variations in the physicochemical parameters were observed at different spots. Observations in the present study have emphasized the need to raise awareness among the people for water conservation and management.

Key words: Essential, Ecosystem, Ponds, water temperature, TDS, pH, dissolved oxygen, BOD

INTRODUCTION

Bihar in general and North Bihar particularly is very rich in water resources.[1] It has a number of water bodies like rivers, streams, ponds, lakes and ditches. Water has always been an important and life sustaining drink to humans and it is essential for the survival of all organisms. North Bihar has various ponds. These ponds are very significant for the people of Vaishali district specially for the area where these ponds are situated. Many researchers have worked in Physico-chemical characteristics of ponds in India.[2-7] The present investigation was carried out to evaluate the physico-chemical characteristics of water of ponds named Taj baj Pokhra situated at Hajipur town and Rani Pokhar at village Rani Pokhar of Vaishali district of Bihar.

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MATERIALS & METHODS

Pond 1 (Taj Baj Pokhra) Pond 2 (Rani Pokhar) were selected for monthly sampling. Samples were collected for a period of one year (March, 2022 to February 23). The sampling period was divided in three seasons - summer (March to June), Rainy (July to October) and winter (November to February). The sampling was done during morning (9-10 am). The water samples for physico-chemical analysis were collected in five litre plastic jars from each site. The selected physico-chemical parameters analyzed during investigation were - temperature, turbidity,

pH, total hardness, total dissolved solids, total alkalinity, dissolved oxygen, sulphate and chloride. Selected physico-chemical parameters were analysed with the help of standard analytical methods.

RESULTS & DISCUSSION

Results of the monthly observations of present investigation was arranged for different seasons viz.

Water Temperature

The water temperature ranged between 16.3 - 31.3°C for Pond 1 and 16.7-32.4°C for Pond 2. The maximum temperature was recorded during summer followed by rainy season while the minimum temperature was observed during winter months. It was observed that water temperature is influenced by air temperature. Water temperature influences the level of oxygen of water body. In summer months water temperature is generally higher as compared to winter months.

Turbidity

The maximum value of turbidity was during rainy season and minimum during winter. It ranged between 20.64 to 30.8 for Pond 1 and 21.1 to 32.7 for Pond 2. The seasonal variation in turbidity value was very similar to findings of Varun Prasadh and Daniel (2010)[8]. Higher turbidity in river water might be due to growth of phytoplankton, human activities etc.

pH

pH is one of the most important parameters in water chemistry. It also suggests that whether the water is suitable for drinking or not. The pH ranged between 6.9 - 7.10 for Pond 1 and 6.32 - 7.23 for Pond 2. The data of the present study reflected highest pH value during winter when the water is slightly alkaline. These findings are also accordance with Mishra *et al.* (2008)[9]. The pH value of water during summer and rainy season is very close but a little less than 7. The microbiological integrity of water also depends upon its pH value.

Total Hardness

The total hardness of water is caused by carbonates, bicarbonates, sulphate, chlorides and nitrates of calcium and magnesium ions. In present investigation maximum value of total hardness was observed during winter season, followed by rainy season. The total hardness of water ranged between 185-240 for Pond 1 and 194 - 245 for Pond 2. Present investigation suggests that water of every site in all seasons is hard. Water with a hardness upto 75 mg/l is treated as soft, from 75 - 150 mg/l moderately hard, 150-300 mg/l as hard and above that very hard.[10,11] Total hardness values of river water at all the three sites in different seasons were found to be well within the BIS permissible limit of 500 mg/l.

Total Dissolved Solids

Dissolved solid substances influence the taste, hardness and corrosive property of water. Dissolved solids in water include organic salts and small amount of organic matter. In present investigation, the amount of dissolved solids ranged between 410 - 434 for Pond 1 and 419- 444 for Pond 2. The high amount of suspended, dissolved and total solids adversely affects the quality of water and unsuitable for any purpose including irrigation.

Total Alkalinity

Total alkalinity of water is due to the presence of mineral salts. It is due to carbonate and bicarbonate ions. Total alkalinity of water ranged between 32.2 - 50.23 for Pond 1 and 31.8 - 53.10 for Pond 2. The maximum alkalinity was observed during summer and minimum during winter season. The low alkalinity during winter is due to dilution. Alkalinity is directly related to the productivity of water bodies because it regulates the pH and free CO_2 of the water bodies.

BOD

Biochemical oxygen demand (BOD) is one of the most important parameters as it reflects status of aquatic pollution. BOD is the oxidisable organic matter found in water and its value may be used as a measure of waste strength.[11] BOD ranged between 7.3 - 8.8 for Pond 1 and 6.9 - 8.6 for Pond 2. BOD was recorded maximum during summer while minimum during winter. The data of BOD suggests that ponds were moderately polluted since water having BOD less than 1.0 mg/l is unpolluted between 2.0 - 9.0 mg/l is moderately polluted and above 9.0 mg/l is heavily polluted.

Dissolved Oxygen

Dissolved oxygen is an important parameter for water quality. It also serves as an indicator of the physical, chemical and biological activities of the water body. During investigation the minimum value of DO_2 of water was recorded during summer months while maximum during winter. DO_2 ranged between 6.32 - 7.80 Pond 1 and 6.23 - 7.50 for Pond 2. The highest dissolved oxygen during winter months might be due to high photosynthetic activity during these months.

Sulphate

The maximum concentration of sulphate was recorded during winter while minimum during summer months. It ranged between 139.63 - 185.3 for Pond 1 and 142.50 - 183.96 for Pond 2.

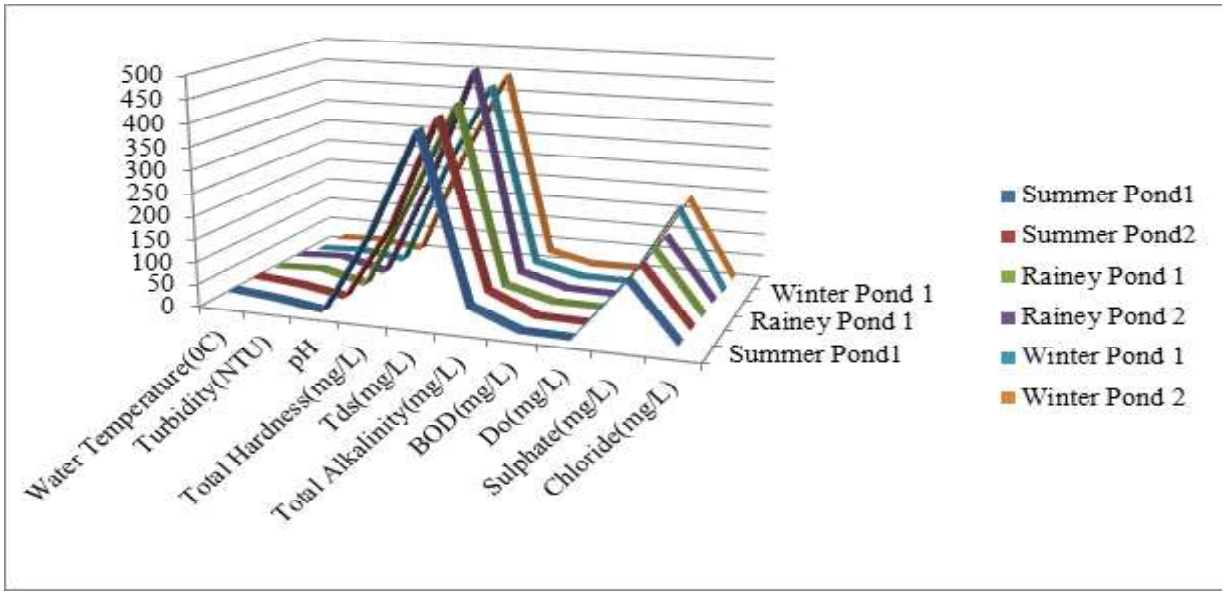
Chloride

The level of chloride ranged between 6.23 - 18.63 for Pond 1, 6.67 - 17.49 for Pond 2. It is evident here that chloride was minimum during winter and maximum during summer season. The major anthropogenic sources of chloride in surface waters include dicing salt, urban an agriculture run off, municipal discharge etc. Chloride is an index of human pollution origin.

Table 1- Showing Physico-Chemical status of two ponds of Vaishali district.

Physico–Chemical Parameters	Summer		Rainy		Winter	
	Pond 1	Pond 2	Pond 1	Pond 2	Pond 1	Pond 2
Water Temperature (ºC)	31.3	32.4	26.6	27.5	16.3	16.7
Turbidity (NTU)	21.65	22.89	30.8	32.7	20.64	21.1
pH	6.9	6.85	6.12	6.32	7.10	7.23
Total Hardness (mg/L)	185	175	210	225	240	245
Tds (mg/L)	410	419	429	486	434	444
Total Alkalinity(mg/L)	50.23	53.10	36.3	36.7	32.2	31.8
BOD(mg/L)	8.7	8.3	8.8	8.6	7.3	6.9
Do(mg/L)	6.36	6.29	7.80	6.23	6.32	7.5
Sulphate(mg/L)	139.63	142.50	145.89	147.30	185.3	183.96
Chloride(mg/L)	18.63	17.49	13.54	13.33	6.23	6.67

Graph 1- Showing Physico-Chemical status of two ponds of Vaishali district



CONCLUSION

In Bihar, Specially in Vaishali district such study on the pattern of physio-chemical factors of these two different ponds are very few. It is apparently clear that no detailed biological investigations have been done so far in this area. With this objective, the author has studied the comparative account of the Physico-chemical factors and the primary productivity in Taj Baj Pokhra and Rani Pokhar of Vaishali District. This chapter establish the structure of water system, environmental conditions and reflects the health of Taj Baj Pokhra and Rani Pokhar

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